

Data Analysis and Interpretation

ASSIGNMENT - 4

REPORT

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Instructions to run our code

Submission Format

A4-24B1033-24B1059-24B1026/

```
| Question3/  
|   | hw4_q3.m  
| Question6/  
|   | hw4_q6.m  
| report.pdf
```

Detailed Instructions

- First unzip the submission folder.
- To see the results and solutions open report.pdf.
- To view the codes for question 3 open the folder 'Question3'.
- Similarly, to view the code for question 6, open the folder 'Question6'.
- The plots and comments for question 3 and question 6 are given in report.pdf.

1 Question 1

Given a d -dimensional random variable

$$X \sim \mathcal{N}(\mu, C)$$

where $\mu \in \mathbb{R}^d$ and C is a $d \times d$ covariance matrix.
We are to find the PDF of

$$Y = BX + \nu$$

where we assume B and C are invertible matrices.

Mean of Y

$$\begin{aligned} \mu_Y &= \mathbb{E}[Y] \\ &= \mathbb{E}[BX + \nu] \\ &= \mathbb{E}[B]\mathbb{E}[X] + \mathbb{E}[\nu] \\ &= B\mathbb{E}[X] + \nu \quad (\text{since } B \text{ and } \nu \text{ are constants, } \mathbb{E}[B] = B \text{ and } \mathbb{E}[\nu] = \nu) \\ &= B\mu + \nu \quad (\mathbb{E}[X] = \mu \text{ and } \mu \text{ is known to us}) \end{aligned}$$

Hence,

$$\boxed{\mu_Y = B\mu + \nu}$$

Covariance Matrix of Y

$$\begin{aligned} C_Y &= \mathbb{E}[(Y - \mathbb{E}[Y])(Y - \mathbb{E}[Y])^T] \\ &= \mathbb{E}[(BX + \nu - B\mu - \nu)(BX + \nu - B\mu - \nu)^T] \\ &= \mathbb{E}[B(X - \mu)(X - \mu)^T B^T] \\ &= B\mathbb{E}[(X - \mu)(X - \mu)^T] B^T \\ &= BCB^T \end{aligned}$$

Thus,

$$\boxed{C_Y = BCB^T}$$

PDF by Transformation of Variables

We have:

$$Y = BX + \nu \implies X = B^{-1}(Y - \nu)$$

From the change of variables formula:

$$f_Y(y) = \frac{f_X(x)}{|\frac{\partial Y}{\partial X}|}$$

Since $\frac{\partial Y}{\partial X} = B$, we get:

$$f_Y(y) = \frac{f_X(B^{-1}(y - \nu))}{|B|}$$

Substituting PDF of X, Given $X \sim \mathcal{N}(\mu, C)$,

$$f_X(x) = \frac{\exp\left(-\frac{1}{2}(x - \mu)^T C^{-1}(x - \mu)\right)}{(2\pi)^{d/2} |C|^{1/2}}$$

Hence,

$$\begin{aligned} f_Y(y) &= \frac{1}{|B|} \cdot \frac{\exp\left(-\frac{1}{2}(B^{-1}(y - \nu) - \mu)^T C^{-1}(B^{-1}(y - \nu) - \mu)\right)}{(2\pi)^{d/2} |C|^{1/2}} \\ &= \frac{\exp\left(-\frac{1}{2}(y - \nu - B\mu)^T (B^{-1})^T C^{-1} B^{-1}(y - \nu - B\mu)\right)}{(2\pi)^{d/2} |C|^{1/2} |B|} \end{aligned}$$

Noting that:

$$(B^{-1})^T C^{-1} B^{-1} = (BCB^T)^{-1}$$

we can write:

$$f_Y(y) = \frac{\exp\left(-\frac{1}{2}(y - (B\mu + \nu))^T (BCB^T)^{-1}(y - (B\mu + \nu))\right)}{(2\pi)^{d/2} |BCB^T|^{1/2}}$$

Therefore,

$$\boxed{Y \sim \mathcal{N}(B\mu + \nu, BCB^T)}$$

and

$$\boxed{f_Y(y) = \frac{\exp\left(-\frac{1}{2}(y - \nu - B\mu)^T (B^{-1})^T C^{-1} B^{-1}(y - \nu - B\mu)\right)}{(2\pi)^{d/2} |C|^{1/2} |B|}}$$

2 Question 2

$$\hat{C} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T$$

where $\bar{x}$ is the mean of all the vectors $\{x_i\}_{i=1}^n$.

Define

$$z_i = \frac{x_i - \bar{x}}{\sqrt{n-1}} \in \mathbb{R}^d, \quad \text{and} \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Then,

$$\hat{C} = \sum_{i=1}^n z_i z_i^T$$

Let Z be the matrix whose columns are $z_1, z_2, \dots, z_n$:

$$Z = \begin{bmatrix} | & | & \cdots & | \\ z_1 & z_2 & \cdots & z_n \\ | & | & \cdots & | \end{bmatrix} \in \mathbb{R}^{d \times n}$$

Hence, we can write

$$\boxed{\hat{C} = ZZ^T}$$

Now we have,

$$L = Z^T Z \in \mathbb{R}^{n \times n}$$

and

$$\hat{C} = ZZ^T$$

Let v be an eigenvector of L corresponding to eigenvalue μ , i.e.,

$$Lv = \mu v, \quad v \in \mathbb{R}^n, v \neq 0$$

$$Z^T Z v = \mu v$$

Premultiplying both sides by Z gives

$$ZZ^T(Zv) = \mu(Zv)$$

Substituting value of $\hat{C}$

$$\hat{C}(Zv) = \mu(Zv)$$

Thus, the vector

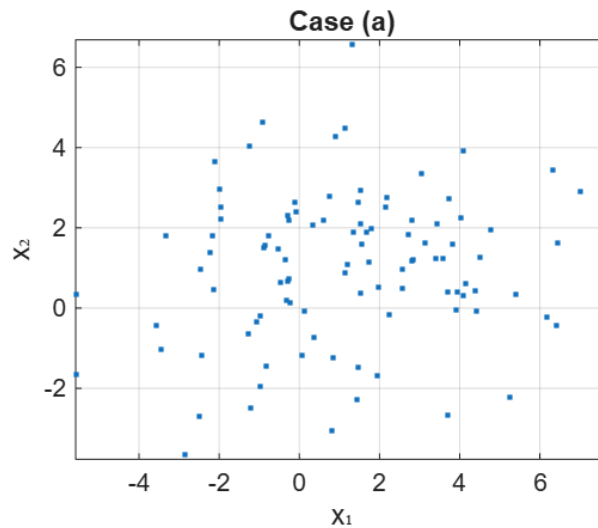
$$u = Zv$$

is an eigenvector of $\hat{C}$ corresponding to eigenvalue $\lambda = \mu$.

The eigenvectors of $\hat{C}$ can be obtained from those of L by multiplying each eigenvector v of L by Z , i.e., $u = Zv$, $\hat{C}u = \lambda u$, $\lambda = \mu$.

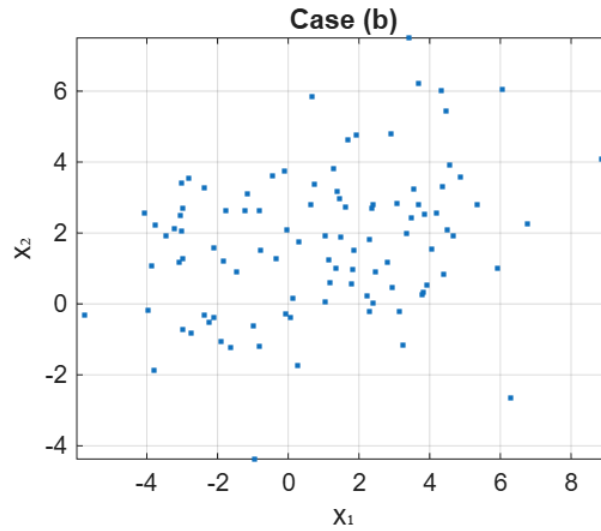
3 Question 3

3.1 Case (a)



Maximum variance direction: $[1.00 \ 0.00]$
 Minimum variance direction: $[0.00 \ 1.00]$

3.2 Case (b)



Maximum variance direction: $[-0.98, -0.19]$
 Minimum variance direction: $[0.19, -0.98]$

4 Question 4

Let $X_1, \dots, X_n$ be samples for $F_n(x)$

$Y_1, \dots, Y_m$ be samples for $G_m(x)$

According to the null hypothesis, all these samples are drawn from $F(x)$.

$$U_i := F(X_i), \quad U_j := F(Y_j)$$

U_i, U_j are uniform random variables $\sim \text{Uniform}(0, 1)$

$$F_n(x) = \frac{\#i : X_i \leq x}{n},$$

$$G_m(x) = \frac{\#j : Y_j \leq x}{m}$$

$$\begin{aligned} E(F_n(x)) &= \frac{1}{n} \sum_{i=1}^n E(\mathbf{1}(X_i \leq x)) \\ &= \frac{1}{n} \sum_{i=1}^n P(X_i \leq x) \\ &= \frac{1}{n} \sum_{i=1}^n F(x) \\ &= F(x) \end{aligned}$$

$$\begin{aligned}
\text{Var}(F_n(x)) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i \leq x)\right) \\
&= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(\mathbf{1}(X_i \leq x)) \\
&= \frac{1}{n^2} \sum_{i=1}^n P(X_i \leq x)(1 - P(X_i \leq x)) \\
&= \frac{n}{n^2} F(x)(1 - F(x)) \\
&= \frac{F(x)(1 - F(x))}{n}
\end{aligned}$$

Similarly,

$$\begin{aligned}
E(G_m(x)) &= F(x) \\
\text{Var}(G_m(x)) &= \frac{F(x)(1 - F(x))}{m}
\end{aligned}$$

$$D_2 = \max_x |F_n(x) - G_m(x)|$$

$$\begin{aligned}
P_F\{D_2 \geq d\} &= P_F\left\{\max_x \left| \frac{\#i : X_i \leq x}{n} - \frac{\#j : Y_j \leq x}{m} \right| \geq d\right\} \\
&\quad (\text{assuming } F(x) \text{ is non-decreasing function}) \\
&= P_F\left\{\max_x \left| \frac{\#i : F(X_i) \leq F(x)}{n} - \frac{\#j : G(Y_j) \leq F(x)}{m} \right| \geq d\right\} \\
&= P\left\{\max_x \left| \frac{\#i : U_i \leq F(x)}{n} - \frac{\#j : U_j \leq F(x)}{m} \right| \geq d\right\}
\end{aligned}$$

Let $F(x) = y \Rightarrow x$ follows distribution F so, $F(x) = u; u \in U(0, 1) \Rightarrow y \in (0, 1)$

$$P_F\{D_2 \geq d\} = P\left\{\max_{0 < y \leq 1} \left| \frac{\#i : U_i \leq y}{n} - \frac{\#j : U_j \leq y}{m} \right| \geq d\right\}$$

Thus, the distribution of D_2 is independent of $F(x)$.

5 Question 5

Let the mean difference be denoted as

$$\mu_d = A - B.$$

We test the null hypothesis:

$$H_0 : \mu_d < 0.$$

The test statistic is given by

$$T = \frac{\bar{d} - \mu_0}{\sigma/\sqrt{n}} \quad (\text{we know that, } \mu_0 = 0).$$

From the given data:

$$\bar{d} = \frac{-7.5 + 2.5 - 2.5 - 3.5 - 1.5 + 0.5 + 1 - 4.5 - 4.5 - 1.5}{10} = -2.15.$$

The sample standard deviation is

$$\sigma_s = \sqrt{\frac{(-7.5 + 2.15)^2 + (2.5 + 2.15)^2 + (-2.5 + 2.15)^2 + (-3.5 + 2.15)^2 + (-1.5 + 2.15)^2 + (0.5 + 2.15)^2 + (1 + 2.15)^2 + (-4.5 + 2.15)^2 + (-4.5 + 2.15)^2 + (-1.5 + 2.15)^2}{9}} \approx 3.$$

Hence, since the sample is symmetric about 0, we have

$$t = \frac{\bar{d} - \mu_0}{\sigma_s/\sqrt{n}} = \frac{(-2.15 - 0)}{3/\sqrt{10}} \approx -2.27.$$

$$P\{T < -2.27\} = P\{T > 2.27\}.$$

The p -value is computed as

$$\begin{aligned} p\text{-value} &= P\{T \geq -2.27\} \\ &= 1 - P\{T < -2.27\} \\ &= 1 - P\{T > 2.27\} \quad (\text{since } T \text{ is symmetric about } 0) \\ &= 1 - 0.025 \quad [\text{from } t\text{-table}] \\ &= 0.975. \end{aligned}$$

Given $\alpha = \frac{5}{100} = 0.05$, we have

$$0.975 > \alpha.$$

Therefore, the null hypothesis is **accepted**, i.e., the average weekly labor hour loss decreases. Hence, the program is **effective**.

6 Question 6

6.1 6(c)

From the graphs below, we can infer the following:

- First few eigenvalues are relatively very high and then we observe a rapid drop.
- After this drop, graph will bend.
- Eigenvalues tend to 0 after that for every digit and graph will become flatter.

6.2 6(d)

From the images, we observed the following:

- The first eigenvector (corresponding to the largest eigenvalue) looks like a blurry version of the digit but is more structured than others.
- We observed that lower the eigenvalue, the corresponding eigenvector gives us a slant digit (either left or right).
- As you look at eigenvectors for smaller eigenvalues, they tend to look less like the digit and capture finer, less obvious details or potentially just noise.

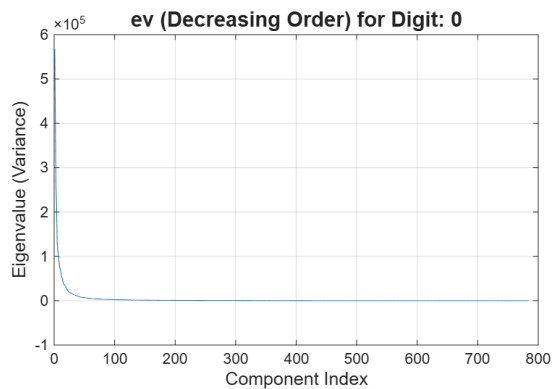


Figure 1: 6(c) Graph of eigenvalue 0

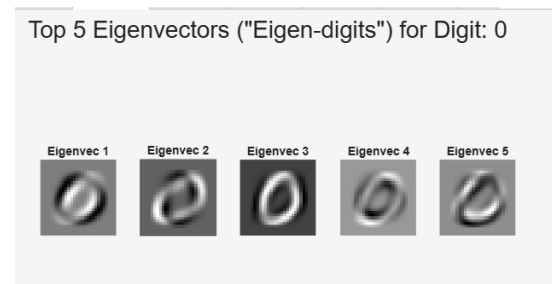


Figure 2: 6(d) Reshaped images of corresponding eigenvectors

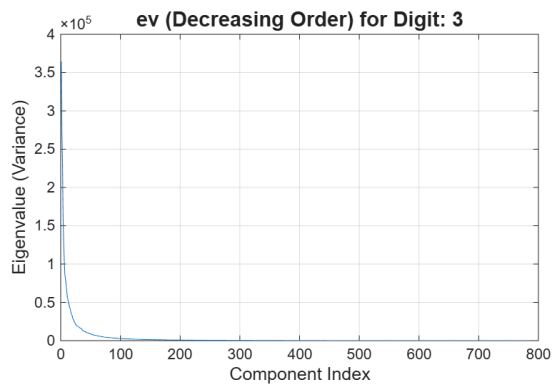


Figure 3: 6(c) Graph of eigenvalue 3

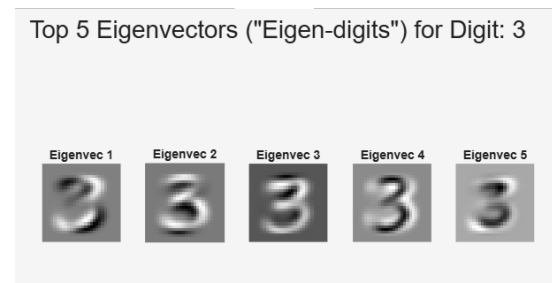


Figure 4: 6(d) Reshaped images of corresponding eigenvectors

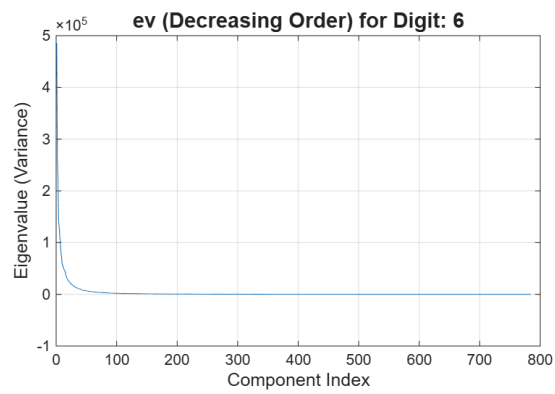


Figure 5: 6(c) Graph of eigenvalue 6

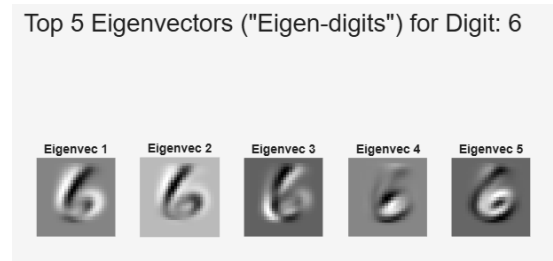


Figure 6: 6(d) Reshaped images of corresponding eigenvectors

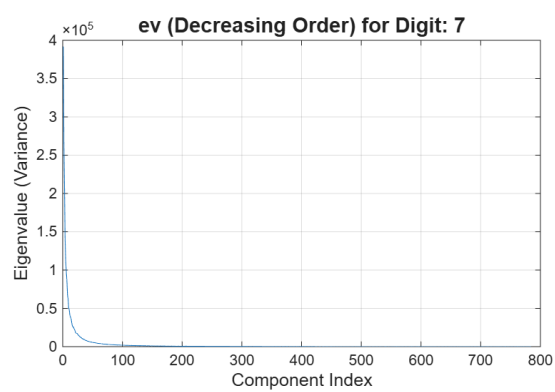


Figure 7: 6(c) Graph of eigenvalue 7

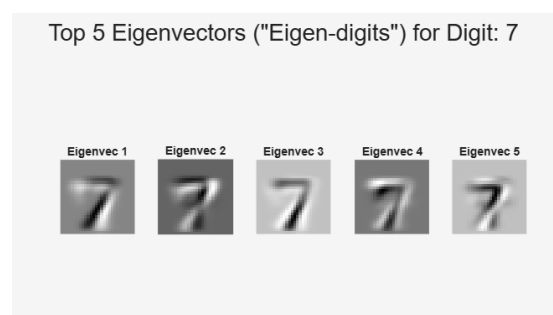


Figure 8: 6(d) Reshaped images of corresponding eigenvectors